

VAMSI ARUMALLA

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SUMMARY

Backend engineer focused on low-latency distributed systems, stream processing, and high-availability cloud services. Currently building the ingestion and decisioning layer of Amazon's real-time payment fraud platform, evaluating 450+ transactions per second (~39M/day) on Java, Kafka, EKS, DynamoDB, and SQS. Re-architected fraud detection from 10-minute batch scoring to sub-5-second streaming, moving intervention ahead of settlement. Deep expertise in event-driven architecture, reliability engineering, and production observability.

EXPERIENCE

Software Engineer | Amazon, Seattle, WA

Feb 2025 – Present

- Own development and production operation of eight Java microservices on EKS that ingest payment events, enrich transaction context, and score fraud risk in real time, operating in the critical path of every evaluated payment.
- Re-architected anomaly detection from delayed batch scoring to a Kafka streaming pipeline, collapsing detection latency from 10+ minutes to under 5 seconds so high-risk transactions are blocked before settlement instead of clawed back after.
- Established SLO-driven reliability across owned services with error budgets, Prometheus/Grafana dashboards, and actionable alerting, cutting incident time-to-detection and helping sustain 99.98% availability.
- Implemented OAuth 2.0/JWT service-to-service authentication across Java and Python APIs, standardizing identity validation between internal services.
- Designed asynchronous event-enrichment middleware that eliminated redundant cross-service validation calls between fraud services.
- Delivered React/TypeScript investigation dashboards that consolidate transaction anomalies, risk signals, and analyst workflows into a single triage interface.

Software Engineer | Capgemini, Tampa, FL

Feb 2024 – Jan 2025

- Developed Spring Boot, Kafka, and PostgreSQL microservices on Azure for high-volume financial transaction processing, exposing REST and GraphQL APIs through OpenAPI-defined contracts.
- Engineered idempotent event processing using retry handling, reconciliation workflows, write-ahead tracking, and Kafka partition tuning, preventing duplicate transaction records during downstream failures.
- Automated release promotion with Docker, Helm, and Azure DevOps pipelines, replacing manual deployment steps with repeatable, environment-specific delivery.

Software Engineer | S7 Works, Hyderabad, India

Jan 2021 – Jul 2022

- Cut inter-service communication failures by 41% across Kubernetes applications through service discovery, network isolation, and fault-containment patterns.
- Introduced CI/CD with Jenkins, Argo CD, and Terraform, enabling automated zero-downtime deployments across 5+ backend services.
- Automated infrastructure provisioning and health-check monitoring, reducing manual operational work and preventing configuration drift between development and production.

SELECTED PROJECT

Distributed Log Ingestion & Query Platform

Kafka, Kubernetes, S3, Prometheus

- Designed a fault-tolerant ingestion platform using partitioned Kafka topics, horizontally scaled stateless processors, backpressure controls, and replayable at-least-once streams validated under injected node failures.
- Implemented hot/cold tiered storage with lag-aware autoscaling, maintaining stable p95 query latency under 10x burst traffic and recovering in under 60 seconds during failover tests.

EDUCATION

M.S. Computer Science | Sacred Heart University, Fairfield, CT

Sep 2022 – Dec 2023

B.Tech. Computer Science | GNA University, Punjab, India

Jun 2018 – Apr 2022

TECHNICAL SKILLS

Languages & Backend: Java, Python, TypeScript/React, SQL, Spring Boot, REST, GraphQL, OpenAPI

Distributed Systems: Kafka, Kafka Streams, event-driven architecture, stream processing, idempotency, distributed systems design

Cloud & Infrastructure: AWS (EKS, Lambda, SQS, S3, CloudWatch), Azure, Kubernetes, Docker, Helm, Terraform

Data Stores: PostgreSQL, DynamoDB, Redis, Elasticsearch

Reliability & Delivery: Prometheus, Grafana, SLOs/SLIs, GitHub Actions, Argo CD, Jenkins, Azure DevOps